

Annotated Paper List: Avidity, Multivalent Binding, and Selective Cell Targeting

Scope: Focused review, 2021–2026
Date compiled: 2026-09-04
Total papers: 76, organized into 6 thematic sections

1. Avidity Theory and Multivalent Binding Models

#	Title	Authors	Year	Journal	Key Finding
1	Binding Revisited—Avidity in Cellular Function and Signaling	Erlendsson, Teilum	2021	Front. Mol. Biosci.	Review defining avidity as accumulated strength of multiple interactions; frames avidity's role in intracellular signaling and receptor clustering. 99 citations.
2	Avidity in antibody effector functions and biotherapeutic drug design	Oostindie et al.	2022	Nat. Rev. Drug Discov.	Landmark review: multivalent binding lowers effective binding threshold and amplifies effector functions; coordinated multivalency more potently triggers receptor clustering than single high-affinity interactions.
3	Divalent ligand-monovalent molecule binding	Janssen, Stenmark, Carlson	2021	bioRxiv	Theoretical: intermediate AA·B2 complex in divalent ligand–monovalent receptor binding governed by cubic equation; quantifies how depletion of both ligand and receptor shapes avidity and effective concentrations.
4	Theoretical Analysis of Bivalent Binding Avidity and Kinetics:	Watts et al.	2025	ChemRxiv	Shortening tether increases equilibrium bivalent avidity by raising local concentration of second

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	Implications for DNA-Encoded Libraries				ligand, but overly short tethers slow target searching—trade-off between avidity and kinetics.
5	Practical guide to calculating impact of avidity in models of bispecific antibodies with two membrane-bound targets	Kareva et al.	2022	bioRxiv	Practical calculation method for avidity effect in bispecific antibodies binding two membrane-bound targets on same cell; shows how relative target concentrations alter free/bound antibody concentrations.
6	Nanoscale spatial dependence of avidity in an IgG1 antibody	Jendroszek, Kjaergaard	2021	Sci. Rep.	IgG1 avidity maximized at ~13 nm inter-epitope spacing; relatively insensitive within 7–20 nm range (~2-fold change). Directly relevant to linker length optimization.
7	Strategies for targeting cell surface proteins using multivalent conjugates and chemical biology	Sachdev, Cabalteja, Cheloha	2021	Methods Cell Biol.	Review of bivalent compounds combining antibody affinity/specificity with synthetic modularity for cell surface targeting.
8	A quantitative view of strategies to engineer cell-selective ligand binding	Tan, Orcutt-Jahns, Meyer	2021	Integr. Biol.	General multivalent binding model showing how receptor density, avidity, valency, and multi-specificity can be tuned—alone or combined—to optimize cell-selective targeting.

2. Receptor Density, Superselectivity, and Targeting Selectivity

#	Title	Authors	Year	Journal	Key Finding
9	Determinants of Superselectivity—Practical Concepts for Application in Biology and Medicine	Dubacheva, Curk, Richter	2023	Acc. Chem. Res.	Introduces scaling variable xS that quantitatively predicts how receptor density, valency, affinity, and competitor effects determine the sharp on/off threshold of superselective multivalent binding.
10	Kinetic Superselectivity in Multivalent Binding	Ravnik et al.	2026	J. Phys. Chem. Lett.	Both association and dissociation can exhibit superselectivity exceeding equilibrium predictions, driven by two-step binding mechanism (fast weak $\rightarrow$ slow strong).
11	Controlling Superselectivity of Multivalent Interactions with Cofactors and Competitors	Curk et al.	2022	JACS	Cofactors and competitors modulate superselectivity by rescaling effective ligand–receptor affinity, enabling quantitative control of receptor–density–dependent avidity.
12	Receptor clustering tunes and sharpens the selectivity of multivalent binding	Xie et al.	2025	PNAS	Tuning inter-receptor attraction enhances or suppresses multivalent binding; receptor clustering regulates selectivity. 14 citations.
13	The Role of Receptor Uniformity in Multivalent Binding	Xia et al.	2023	JACS Au	Receptor uniformity improves selectivity at intermediate binding energies; provides guidelines for biosensor/nanoparticle design.
14	Valency and affinity control of aptamer-conjugated	Woythe et al.	2023	J. Control. Release	High-affinity aptamers + low valency nanoparticles increase selective targeting to high-density cancer

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	nanoparticles for selective cancer cell targeting				cells; reduce non-tumor accumulation. 33 citations.
15	A Single-Molecule View at Nanoparticle Targeting Selectivity: Correlating Ligand Functionality and Cell Receptor Density	Woythe et al.	2022	ACS Nano	dSTORM single-molecule mapping correlates ligand functionality with receptor density for rational design of selective nanomedicines. 71 citations.
16	The quantity of ligand–receptor interactions between nanoparticles and target cells	Schorr et al.	2025	Nanoscale Horiz.	Optimizing ligand number on nanoparticles is crucial for avidity and selectivity; ~251 binding ligands per nanoparticle identified as optimal.
17	Ligand Separation on Nanoconstructs Affects Targeting Selectivity to Protein Dimers on Cell Membranes	Wu et al.	2023	Nano Lett.	Ligand density matching dimer spacing yields higher selectivity for dimeric protein forms on cell membranes.
18	Improving Cancer Targeting: Effect of Dual-Ligand Density on Targeting Cells with Differential Biomarker Expression	Sultana, David	2023	Int. J. Mol. Sci.	Optimal ligand density depends on biomarker expression levels on target cells.

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19	Valency-affinity mapping of multivalent liposomes for tunable target cell discrimination	Garcia et al.	2025	Drug Deliv.	Binding performance index (BPI) increases with receptor density and DARPIn affinity/valency; enables tunable target discrimination.
20	From Single Ligand-Receptor Bond Strength to Collective Avidity: Mechanics-Guided Superselective Nanoparticle Adhesion	Hamzeh et al.	2025	Langmuir	NP mechanics (rigid/semirigid/deformable) determine avidity regime; deformable NPs achieve high avidity at low receptor density via shape adaptation.
21	Single-Molecule Multivalent Interactions Revealed by Plasmon-Enhanced Fluorescence	Okholm et al.	2024	ACS Nano	Trivalent DNA ligands show polynomial avidity increases with receptor density; spacer-length-dependent decrease in binding. Directly quantifies valency-spacer-avidity relationship.
22	Quantifying and controlling bond multivalency for advanced nanoparticle targeting to cells	Makhani et al.	2021	Nano Convergence	Review: nanoparticle avidity depends on ligand density, size/shape, receptor density, monovalent kinetics; multivalent interactions markedly enhance adhesion and internalization.
23	Multivalent Targeting of the Asialoglycoprotein Receptor by Virus-Like Particles	Hincapie et al.	2023	Small	VLPs efficiently target ASGPR for liver-selective trafficking, but barrier cells still take up most particles —highlights selectivity challenge.

3. Bispecific/Trispecific Antibodies and AND-Gate Targeting

#	Title	Authors	Year	Journal	Key Finding
24	Improving dual targeting selectivity in T-cell engagers via synapse-gated and affinity-tuned trispecific antibody design	Zhao et al.	2025	mAbs	Synapse-gated, affinity-tuned trispecific design enhances dual-target selectivity; higher tumor-restricted T-cell activation, reduced on-target/off-tumor activity vs. bispecifics.
25	A logic-gated trispecific engager enhances macrophage killing of cancer cells in solid tumors	Zhao et al.	2026	Nat. Biotechnol.	Trispecific macrophage engager (TrME) activates LRP1 + blocks SIRPα + tumor-targeting arm; mRNA-LNP delivery in vivo inhibits tumor growth and extends survival.
26	Improving the tumor selectivity of T cell engagers by logic-gated dual tumor-targeting	Shen et al.	2023	Pharmacol. Res.	TriTCEs co-recognizing two tumor antigens achieve ~18 pM EC50 cytotoxicity, 70–750× greater activity than single-antigen controls.
27	Development of T-cell engagers selective for cells co-expressing two antigens	Dicara et al.	2022	mAbs	Trispecific T-cell engagers targeting Ly6E + B7-H4 + CD3 kill only when both antigens co-expressed; in vitro clearance correlates with higher-affinity B7-H4 domain.
28	Trispecific T-cell engagers for dual tumor-targeting of colorectal cancer	Tapia-Galisteo et al.	2022	Oncoimmunology	Trispecific T-cell engagers achieve selective dual-tumor targeting with gate-like control over T-cell activation vs. monospecific/bispecific formats.
29	When three is not a crowd: trispecific	Tapia-Galisteo et al.	2023	Theranostics	Review: trispecific antibodies improve selectivity and potency at

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	antibodies for enhanced cancer immunotherapy				lower concentrations vs. bispecifics; co-engaging two tumor antigens + CD3 or NK-activating receptors.
30	Next generation of multispecific antibody engineering	Keri et al.	2023	Antib. Ther.	Review: multispecific design achieves programmable selectivity and gate-like antigen recognition; emphasizes need for new design workflows.
31	Design of Crosslinking Antibodies For T-Cell Activation: PD-1/CD137 Bispecific Agents	Kopp et al.	2024	AAPS J.	Target expression is dominant factor for T-cell activation specificity; affinity tuning and non-overlapping epitope combinations improve activation at low expression levels.
32	A bivalent, bispecific Dab-Fc antibody molecule for dual targeting of HER2 and HER3	Rau et al.	2021	mAbs	Dab-Fc effectively targets HER2/HER3; potent inhibition of tumor proliferation and migration in gastric cancer xenograft.
33	Bispecific DNA-Peptide Probes for Targeting Receptor Pairs on Live Cells	Ghosh et al.	2025	Angew. Chem.	Bispecific DNA-peptide probes target specific receptor pairs at lower molecular weight than previous agents; potential for diagnostics and therapeutics.
34	Engineering a β -Sheet Enables Bispecific Binding in Single VHH Domains	Torres et al.	2026	ACS Synth. Biol.	Single-domain VHH with engineered β -sheet paratope achieves bispecific binding within one scaffold; modularity for high-affinity targeting.

4. Nanobody Multimerization, Avidity, and Linker Design

#	Title	Authors	Year	Journal	Key Finding
35	Multivalent CXCR4-targeting nanobody formats differently affect affinity, receptor clustering, and antagonism	Anbuhl et al.	2024	Biochem. Pharmacol.	Fc-tail fusion in multivalent CXCR4 nanobodies yields highest avidity effect; increased potency in inhibiting CXCL12-mediated signaling. 13 citations.
36	Advances in nanobody multimerization and multispecificity: from in vivo assembly to in vitro production	Al-Seragi et al.	2025	Biochem. Soc. Trans.	Review: multimerization via flexible linkers and self-assembling modules modulates avidity and kinetics; gains depend on linker length, valency, and geometry.
37	Nanobodies: Robust miniprotein binders in biomedicine	Kim et al.	2023	Adv. Drug Deliv. Rev.	Review: multivalent Nb constructs show markedly enhanced avidity and slower dissociation vs. monovalent; gains strongly depend on linker length for cooperative binding.
38	Structure-guided design of a trivalent nanobody cluster targeting SARS-CoV-2 spike protein	Jiang et al.	2024	Int. J. Biol. Macromol.	Trivalent "Tribody" nanobody cluster achieves 900-fold increase in neutralization vs. monovalent using self-assembled scaffolds.
39	Anti-HIV-1 Nanobody-IgG1 Constructs With Improved	Schriek et al.	2022	Front. Immunol.	Bivalent/trivalent nanobody constructs with long glycine-serine linkers show

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	Neutralization Potency and Fc Effector Functions				increased HIV-1 Env binding and enhanced neutralization vs. monovalent.
40	Nanobody assemblies with fully flexible topology enabled by genetically encoded tetrazine amino acids	Van Fossen et al.	2022	Sci. Adv.	Tetrazine-conjugated nanobody dimers/trimers show improved neutralization and binding; topology and linker placement tune avidity.
41	Plug-and-play nucleic acid-mediated multimerization of biparatopic nanobodies for molecular imaging	Teodori et al.	2024	Mol. Ther. Nucleic Acids	Holliday junction scaffold multimerizes biparatopic Nbs with enhanced avidity; clear linker configuration dependence.
42	Linker-optimized bivalent nanobodies combined with photothermal gold liposome nanocomposites for enhanced dual-mode lateral flow immunoassay	Han et al.	2026	Biosens. Bioelectron.	Linker-optimized bivalent nanobodies enhance immunoassay sensitivity 10.7-fold; demonstrates linker length optimization impact.
43	Affinity Enhancement in Discrete Multivalent MegaMolecules	Gu et al.	2026	ChemBioChem	Multivalent megamolecules with targeted nanobodies enhance binding affinities to EGFR/HER2.

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44	Enhancing the sensitivity of nanobodies through covalent and non-covalent polymerization	Wang et al.	2026	RSC Adv.	Covalent dimerization + non-covalent heptamerization yields 14-mer nanobody with >10-fold sensitivity improvement (SPR-detected avidity).
45	Enhancing Nanobody Immunoassays through Ferritin Fusion: Salmonella-Specific Fenobody	Liao et al.	2024	J. Agric. Food Chem.	Ferritin fusion significantly improves avidity and sensitivity of nanobody immunoassays. 24 citations.
46	Enhanced sandwich immunoassay based on bivalent nanobody for foodborne pathogens detection	Liao et al.	2024	Anal. Chim. Acta	Bivalent nanobodies with improved affinity and reduced spatial orientation enhance immunoassay performance. 38 citations.
47	Structural and Computational Studies of SARS-CoV-2 Spike Protein Binding with Nanobodies: Avidity-Driven Engineering	Verkhivker	2022	Int. J. Mol. Sci.	Review: nanobodies exploit distinct epitopes and avidity/allosteric cooperativity for mutation-resistant therapies.
48	Biolayer interferometry for measuring kinetics of protein-protein interactions and nanobody binding	Bates et al.	2024	Nat. Protoc.	Protocol for high-throughput kinetic measurements of nanobody binding via BLI. 54 citations.

5. Structure Prediction Metrics (ipTM, pDockQ, ipSAE) and Binding Affinity Correlation

#	Title	Authors	Year	Journal	Key Finding
49	Rēs ipSAE loquuntur: What's wrong with AlphaFold's ipTM score and how to fix it	Dunbrack	2025	bioRxiv	ipSAE score outperforms ipTM in predicting protein-protein interactions and separating true from false complexes, even with disorder or accessory domains. 74 citations.
50	Predicting Experimental Success in De Novo Binder Design: Meta-Analysis of 3,766 Binders	Overath et al.	2025	bioRxiv	AF3-derived ipSAE outperforms commonly used scores (ipTM, pDockQ) in predicting experimental success in de novo binder design across diverse targets. 23 citations.
51	Assessing scoring metrics for AlphaFold2 and AlphaFold3 protein complex predictions	Genz et al.	2025	Protein Sci.	Interface-specific metrics ipTM and pDockQ/ipSAE-based scores best discriminate correct from incorrect complex predictions; ColabFold without templates yields strongest correlation.
52	Analysing open-source protein folding models for nanobody binding prediction	Vogt et al.	2026	Front. Bioinform.	No single metric strongly correlates with binding affinity; Boltz-2 confidence measures best identify true binders; local pLDDT/interface scores best detect OOD sequences and mutations.
53	Evaluation of AlphaFold 3's Protein-Protein Complexes for Predicting Binding Free Energy	Wee, Wei	2024	J. Chem. Inf. Model.	AF3 predicts complexes well but has 8.6% increased error vs. PDB structures; not reliable for flexible regions. 78 citations.

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	Changes upon Mutation				
54	Improved prediction of protein-protein interactions using AlphaFold2	Bryant et al.	2022	Nat. Commun.	AF2 identifies 51% of interacting pairs at 1% FPR for heterodimeric complexes. 966 citations.
55	Evaluation of AlphaFold-Multimer prediction on multi-chain protein complexes	Zhu et al.	2023	Bioinformatics	AF-Multimer promising for large complexes but needs improved multi-interface evaluation and new interface quality scores. 192 citations.
56	A Folding-Docking-Affinity framework for protein-ligand binding affinity prediction	Wu et al.	2025	Commun. Chem.	FDA framework predicts protein-ligand binding affinity using 3D structures. 22 citations.
57	Protein-ligand binding affinity prediction using multi-instance learning with docking structures	Kim et al.	2025	Front. Pharmacol.	Multiple docking poses predict affinity without requiring co-crystal structures.
58	Systematic Evaluation of AlphaFold2 and OpenFold3 on Protein-Peptide Complexes	Fayetorbay et al.	2026	bioRxiv	AF2 outperforms OpenFold3 for protein-peptide complexes; peptide composition and length modulate success.
59	AlphaFold models of host-pathogen interactions elucidate molecular mimicry	Baptista et al.	2025	bioRxiv	AF2/3 yield high-confidence models for 803/6,782 pathogen-human PPIs; pDockQ/ipSAE confidence correlates with accurate binding interface

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					prediction (8/12 validated).

6. Targeted Delivery: AAV, LNP, and Extracellular Vesicles with Multivalent Binders

#	Title	Authors	Year	Journal	Key Finding
60	Balancing Multivalent Avidity and Receptor Availability Governs mRNA Delivery by Antibody-Functionalized LNPs	Hashiba et al.	2026	Nano Lett.	Optimal surface avidity of 0.1 VHH per 100 nm ² balances multivalent engagement with receptor preservation; enables selective mRNA expression in CD8+ T cells and CAR-T generation. Directly addresses avidity–selectivity trade-off in delivery.
61	DNA-directed assembly of multivalent lipid nanoparticles for targeted T cell gene delivery	Kelly et al.	2026	J. Control. Release	Bispecific DNA-directed LNPs substantially improve T cell targeting and transfection vs. monotargeted LNPs; enhanced mRNA expression and altered biodistribution.
62	Secondary Site Ligand for Integrin αVβ3 Enables Targeted mRNA Delivery	Bayer et al.	2026	Angew. Chem.	Multivalent low-affinity non-orthosteric ligands on LNPs yield valency-dependent uptake; 6-fold higher protein expression than controls.
63	Extrahepatic, cell-specific delivery of LNPs through competitive	Bufton et al.	2026	bioRxiv	NanoPilot-coated LNPs redirect delivery from liver to T cells; 30–40% transfection in

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	inhibition of ApoE-mediated uptake				splenic/hepatic T cells, ~3-fold reduced liver accumulation.
64	NEO-AAV: engineered extracellular vesicle-enveloped AAV for activation-coupled T cell transduction and CAR-T generation	Xie et al.	2026	bioRxiv	EV-enveloped AAV with CD7/CD3/CD28 tri-chimera tag achieves single-step T cell transduction; improved antibody-neutralization resistance.
65	Distinguishing Pseudotransduction and True Transduction in EV-AAV Vectors	Boucher et al.	2026	J. Extracell. Vesicles	Uptake-enhancing modifications affect pseudotransduction in actively endocytic cells but increase true transduction in less endocytic cells (e.g., Jurkat T cells).
66	Barcoded Hybrids of Extracellular Vesicles and Lipid Nanoparticles for Multiplexed Tissue Distribution Analysis	Ivanova et al.	2025	Adv. Sci.	Barcoded EV-LNP hybrids enable rapid multiplexed biodistribution profiling across cell sources. 19 citations.
67	Covalent conjugation of extracellular vesicles with peptides and nanobodies for targeted therapeutic delivery	Pham et al.	2021	J. Extracell. Vesicles	Covalent EV conjugation with nanobodies/peptides enables specific delivery to cancer cells without genetic/chemical modification. 207 citations.

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68	Targeted delivery of extracellular vesicles: mechanisms, techniques and therapeutic applications	Zhao et al.	2024	Mol. Biomed.	Review: engineered EVs with enhanced targeting promote therapeutic efficacy. 72 citations.
69	Rational design of extracellular vesicles for targeted drug delivery across physiological barriers	Baby et al.	2026	Nano Today	Review: EVs require tissue-specific targeting strategies to overcome biological barriers. 16 citations.
70	Targeted therapy using engineered extracellular vesicles: principles and strategies for membrane modification	Liu et al.	2023	J. Nanobiotechnol.	Review: surface modifications of EVs significantly improve targeted delivery efficiency. 178 citations.
71	Extracellular vesicles as a next-generation drug delivery platform	Herrmann et al.	2021	Nat. Nanotechnol.	Review: EVs offer advantages over synthetic carriers but need further engineering for clinical translation. 1,650 citations.
72	Extracellular Vesicle-Encapsulated AAVs for Therapeutic Gene Delivery to the Heart	La Salvia et al.	2022	Circulation	EV-AAVs show higher potency and efficacy for cardiac gene delivery vs. free AAVs. 58 citations.
73	Extracellular vesicles as vehicles for drug delivery to the heart	Ilahibaks et al.	2024	Eur. Heart J.	Review: EVs as safer, more effective vehicle for heart disease targeted delivery.
74	Tissue-specific gene delivery approaches	Nasr et al.	2026	Bioeng. Transl. Med.	Review: AAVs, LNPs, polymeric NPs, and EVs all rely on

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					engineered structure-function relationships and high-throughput screening for tissue-specific delivery.
75	Comparative analysis of nucleic acid delivery systems for gene therapy	Saad	2024	Nucleic Acid Insights	EVs and LNPs show distinct delivery efficiencies and immunogenicity; EVs potentially safer multivalent platform vs. AAV/LNP.

Cross-Cutting Themes

- 1. **Avidity–selectivity trade-off:** Multiple papers (especially [60], [9], [14], [19]) show that increasing valency/avidity improves on-target binding but can saturate receptors and reduce selectivity. Optimal avidity balances multivalent engagement with receptor preservation.
- 2. **Linker length as a design parameter:** Papers [4], [6], [36], [37], [42] demonstrate that linker length directly controls effective concentration of the second binding event, with optimal ranges (e.g., ~13 nm inter-epitope spacing for IgG1 [6]) and trade-offs between equilibrium avidity and binding kinetics [4].
- 3. **Receptor density as selectivity driver:** Superselectivity theory ([9], [10], [11]) provides the quantitative framework: multivalent binding creates sharp on/off thresholds based on receptor density, enabling discrimination between cells with different surface antigen levels.
- 4. **AND-gate logic via multispecificity:** Trispecific designs ([24], [26], [27]) achieve gate-like targeting by requiring co-expression of two antigens, with 70–750× improvement over single-antigen controls [26].
- 5. **Structure prediction metrics for binder design:** ipSAE ([49], [50]) is emerging as superior to ipTM/pDockQ for predicting binder success, but no single metric strongly correlates with absolute binding affinity [52].
- 6. **Delivery platform convergence:** AAV, LNP, and EV platforms are converging on multivalent surface display strategies ([60], [61], [67]), with avidity tuning now

being applied at the delivery vehicle level, not just the binder level.